

Matrix Sampling

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1 Notation

$|| \cdot ||$ represents the 2-norm.

$\lambda_{\max}(M)$ represents the maximum eigenvalue of the matrix M . Note: $\lambda_{\max}(M) = ||M||$, if M is PSD

M_{ij} represents the element in the i 'th row, j 'th column of the matrix M .

m_i represents the i 'th row of the matrix M .

v_i represents the i 'th entry in the vector v .

$\mathbb{E}$ represents expected value.

$\Pr(E)$ represents the probability that the random event E occurs.

$\mathbf{E}_{ij}$ represents the standard basis matrix, it has a one in the (i, j) position and zeros elsewhere

If $v \in \mathbb{R}^n$, $\text{diag}(v)$ represents an $n \times n$ matrix, with the entry values of the vector v along its diagonal

2 Assumptions

1. $\max_{i \in [n]} ||a_i|| = 1$, i.e. $\forall i, ||a_i|| \leq 1$. Any nonzero matrix can be rescaled s.t. $\max_{i \in [n]} ||a_i|| = 1$.
2. For all $i \in [n]$, $||a_i|| > 0$ (TODO: this doesn't have to be an assumption if we let $Y_i = \begin{cases} \frac{X_i}{||a_i||} & \text{when } ||a_i|| > 0 \\ 0 & \text{when } ||a_i|| = 0 \end{cases}$, just adds unnecessary complexity).
3. The row-magnitudes have a power-law distribution, i.e. if we assume without loss of generality that the rows of A are ordered by decreasing $||a_i||$, and satisfy $||a_i|| \leq ci^{-k}$, for $c \geq 0$ and $k \in (\frac{1}{3}, 1) \cup (1, \infty)$. A note on the $\frac{1}{3}$ term: this is a relatively slowly-decreasing power-law distribution, which allows for many non-uniform matrices to fall within this requirement, while still resulting in strong concentration measures (to be proven).

3 The Sampling Technique

$A \in \mathbb{R}^{n \times n}$, and $Y \in \mathbb{R}^n$, and let $N \in \mathbb{Z}^+$ be the number of independently generated approximations of A , i.e. for all $i \in [N]$,

$$\tilde{A}_i = D_i A = \text{diag}(y_i) A = \sum_{j=1}^n \frac{X_{ij}}{\sqrt{||a_j||}} \mathbf{E}_{jj} A$$

$$\tilde{A}_i^T \tilde{A}_i = A^T D_i^2 A = A^T \text{diag}(y_i)^2 A = \sum_{j=1}^n \frac{X_{ij}}{\|a_j\|} a_j^T a_j$$

Where for all $i \in [N]$, and $j \in [n]$ $Y_{ij} = \frac{X_{ij}}{\sqrt{\|a_j\|}}$, where $X, Y \in \mathbb{R}^{N \times n}$ and for all $i \in [N]$, and $j \in [n]$, the random variables X_{ij} are svd Bernoulli variables and $X_{ij} = \begin{cases} 1 & \text{w.p. } \|a_j\| \\ 0 & \text{w.p. } 1 - \|a_j\| \end{cases}$.

NOTE: $\mathbb{E}[X_i] = \|a_i\|$

Let...

1. $Z = \frac{1}{N} \sum_{i=1}^N \tilde{A}_i^T \tilde{A}_i - A^T A$, $Z \in \mathbb{R}^{n \times n}$
2. $Z_{ij} = (\frac{X_{ij}}{\|a_j\|} - 1) \frac{a_j^T a_j}{N}$, $Z_{ij} \in \mathbb{R}^{n \times n}$
3. w be s.t. $w^T (A^T A) w = \lambda_{\max}(A^T A)$
4. $\tilde{w}$ be s.t. $\tilde{w}^T (\frac{1}{N} \sum_{i=1}^N \tilde{A}_i^T \tilde{A}_i) \tilde{w} = \lambda_{\max}(\frac{1}{N} \sum_{i=1}^N \tilde{A}_i^T \tilde{A}_i)$

4 Theory

The goal of this section is to get with high probability, $\|A\tilde{w}\| \in [(1 - \epsilon)\|Aw\|, \|Aw\|]$

Lemma 4.1. $Z = \sum_{i=1}^N \sum_{j=1}^n Z_{ij}$

Proof.

$$\begin{aligned} Z &= \frac{1}{N} \sum_{i=1}^N \tilde{A}_i^T \tilde{A}_i - A^T A \\ &= \frac{1}{N} \sum_{i=1}^N (\tilde{A}_i^T \tilde{A}_i - A^T A) \\ &= \frac{1}{N} \sum_{i=1}^N \sum_{j=1}^n (\frac{X_{ij}}{\|a_j\|} a_j^T a_j - a_j^T a_j) \\ &= \sum_{i=1}^N \sum_{j=1}^n (\frac{X_{ij}}{\|a_j\|} - 1) \frac{a_j^T a_j}{N} \\ &= \sum_{i=1}^N \sum_{j=1}^n Z_{ij} \end{aligned}$$

□

Lemma 4.2. For all $A, B \in \mathbb{R}^{n \times n}$, if A and B are Hermitian, then $\lambda_{\max}(A) - \lambda_{\max}(B) \leq \lambda_{\max}(A - B)$

Proof.

$$\begin{aligned}
\lambda_{\max}(B + (A - B)) &\leq \lambda_{\max}(B) + \lambda_{\max}(A - B) \\
&\iff \lambda_{\max}(A) \leq \lambda_{\max}(B) + \lambda_{\max}(A - B) \\
&\iff \lambda_{\max}(A) - \lambda_{\max}(B) \leq \lambda_{\max}(A - B)
\end{aligned}$$

Notice that the first line holds due to Weyl's Inequality. □

Lemma 4.3. $\lambda_{\max}(A^T A) - \lambda_{\max}(\frac{1}{N} \sum_{i=1}^N \tilde{A}_i^T \tilde{A}_i) \leq \|Z\|$

Proof.

$$\begin{aligned}
\lambda_{\max}(A^T A) - \lambda_{\max}(\frac{1}{N} \sum_{i=1}^N \tilde{A}_i^T \tilde{A}_i) &\leq \lambda_{\max}(A^T A - \frac{1}{N} \sum_{i=1}^N \tilde{A}_i^T \tilde{A}_i) && \textbf{Lemma 4.2} \\
&\leq \|A^T A - \frac{1}{N} \sum_{i=1}^N \tilde{A}_i^T \tilde{A}_i\| \\
&= \|\frac{1}{N} \sum_{i=1}^N \tilde{A}_i^T \tilde{A}_i - A^T A\| \\
&= \|Z\|
\end{aligned}$$

□

Lemma 4.4. $\tilde{w}^T Z \tilde{w} \leq \|Z\|$

Proof. Follows directly from definition of operator norm. □

Lemma 4.5. $\left| \|A\tilde{w}\|^2 - \|Aw\|^2 \right| \leq 2\|Z\|$

Proof.

$$\begin{aligned}
\left| \|A\tilde{w}\|^2 - \|Aw\|^2 \right| &= \left| \tilde{w}^T A^T A \tilde{w} - \lambda_{\max}(A^T A) \right| \\
&= \lambda_{\max}(A^T A) - \tilde{w}^T A^T A \tilde{w} \\
&= \lambda_{\max}(A^T A) - \lambda_{\max}(\frac{1}{N} \sum_{i=1}^N \tilde{A}_i^T \tilde{A}_i) + \lambda_{\max}(\frac{1}{N} \sum_{i=1}^N \tilde{A}_i^T \tilde{A}_i) - \tilde{w}^T A^T A \tilde{w} \\
&= \lambda_{\max}(A^T A) - \lambda_{\max}(\frac{1}{N} \sum_{i=1}^N \tilde{A}_i^T \tilde{A}_i) + \tilde{w}^T (\frac{1}{N} \sum_{i=1}^N \tilde{A}_i^T \tilde{A}_i) \tilde{w} - \tilde{w}^T A^T A \tilde{w} \\
&= \lambda_{\max}(A^T A) - \lambda_{\max}(\frac{1}{N} \sum_{i=1}^N \tilde{A}_i^T \tilde{A}_i) + \tilde{w}^T (\frac{1}{N} \sum_{i=1}^N \tilde{A}_i^T \tilde{A}_i - A^T A) \tilde{w} \\
&= \lambda_{\max}(A^T A) - \lambda_{\max}(\frac{1}{N} \sum_{i=1}^N \tilde{A}_i^T \tilde{A}_i) + \tilde{w}^T (Z) \tilde{w} \\
&\leq \|Z\| + \|Z\| && \textbf{Lemmas 4.2, 4.4} \\
&= 2\|Z\|
\end{aligned}$$

□

Lemma 4.6. For all $i \in [N]$, for all $j \in [n]$, $\mathbb{E}[Z_{ij}] = 0$

Notice that this is the expected value of the summands of Z .

Proof.

$$\begin{aligned}
\mathbb{E}[Z_{ij}] &= \mathbb{E}\left[\left(\frac{X_{ij}}{\|a_j\|} - 1\right) \frac{a_j^T a_j}{N}\right] \\
&= \left(\frac{\mathbb{E}[X_{ij}]}{\|a_j\|} - 1\right) \frac{a_j^T a_j}{N} \\
&= \left(\frac{\|a_j\|}{\|a_j\|} - 1\right) \frac{a_j^T a_j}{N} \\
&= (1 - 1) \frac{a_j^T a_j}{N} \\
&= 0
\end{aligned}$$

□

Lemma 4.7. For all $i \in [N]$, for all $j \in [n]$, $\|Z_{ij}\| \leq \frac{1}{N}$

Notice that this is an upper-bound on the summands of Z .

Proof.

$$\begin{aligned}
\|Z_{ij}\| &= \left\| \left(\frac{X_{ij}}{\|a_j\|} - 1\right) \frac{a_j^T a_j}{N} \right\| \\
&= \frac{1}{N} \left| \frac{X_{ij}}{\|a_j\|} - 1 \right| \cdot \|a_j^T a_j\| \\
&= \frac{1}{N} \left| \frac{X_{ij}}{\|a_j\|} - 1 \right| \cdot \|a_j\|^2 \\
&= \frac{1}{N} |X_{ij} \|a_j\| - \|a_j\|^2| \\
&= \begin{cases} (\|a_j\| - \|a_j\|^2)/N & \text{when, } X_{ij} = 1 \\ \|a_j\|^2/N & \text{when, } X_{ij} = 0 \end{cases} \\
&\leq \begin{cases} 1/2N & \text{when, } X_{ij} = 1 \\ 1/N & \text{when, } X_{ij} = 0 \end{cases} \\
&\leq \frac{1}{N}
\end{aligned}$$

$$x \in \mathbb{R}^{n \times 1} \implies \|x^T x\| = \|x\|^2$$

$$x \in [0, 1] \implies x - x^2 \in [0, \frac{1}{2}]$$

□

Lemma 4.8. For all $N > 0$, $\left\| \sum_{i=1}^N \sum_{j=1}^n \mathbb{E}[Z_{ij}^2] \right\| = \frac{1}{N} \left\| \sum_{i=1}^n (\|a_i\| - \|a_i\|^2) a_i^T a_i \right\|$

Notice that this is "the operator norm of the matrix variance of the sum"

Proof.

$$\begin{aligned}
\left\| \sum_{i=1}^N \sum_{j=1}^n \mathbb{E}[Z_{ij}^2] \right\| &= \left\| \sum_{i=1}^N \sum_{j=1}^n \mathbb{E} \left[\left(\frac{X_{ij}}{\|a_j\|} - 1 \right) \frac{a_j^T a_j}{N} \right]^2 \right\| \\
&= \left\| \sum_{i=1}^N \sum_{j=1}^n \mathbb{E} \left[\frac{X_{ij}}{\|a_j\|^2} - 2 \frac{X_{ij}}{\|a_j\|} + 1 \right] \frac{a_j^T a_j a_j^T a_j}{N^2} \right\| \\
&= \left\| \sum_{i=1}^N \sum_{j=1}^n \left(\frac{\mathbb{E}[X_{ij}]}{\|a_j\|^2} - 2 \frac{\mathbb{E}[X_{ij}]}{\|a_j\|} + 1 \right) \frac{\|a_j\|^2 a_j^T a_j}{N^2} \right\| & \forall x \in \mathbb{R}^{1 \times n}, xx^T = \|x\|^2 \\
&= \left\| \sum_{i=1}^N \sum_{j=1}^n \left(\frac{1}{\|a_j\|} - 1 \right) \frac{\|a_j\|^2 a_j^T a_j}{N^2} \right\| \\
&= \frac{1}{N} \left\| \frac{1}{N} \sum_{i=1}^N \sum_{j=1}^n (\|a_j\| - \|a_j\|^2) a_j^T a_j \right\| \\
&= \frac{1}{N} \left\| \sum_{i=1}^n (\|a_i\| - \|a_i\|^2) a_i^T a_i \right\|
\end{aligned}$$

□

Lemma 4.9. For all $N > 0$, $\| \sum_{i=1}^N \sum_{j=1}^n \mathbb{E}[Z_{ij}^2] \| \leq \frac{1}{4N} \|Aw\|^2$

Notice that this is "the operator norm of the matrix variance of the sum"

Proof.

$$\begin{aligned}
\| \sum_{i=1}^N \sum_{j=1}^n \mathbb{E}[Z_{ij}^2] \| &= \frac{1}{N} \| \sum_{i=1}^n (\|a_i\| - \|a_i\|^2) a_i^T a_i \| & \textbf{Lemma 4.8} \\
&\leq \frac{1}{4N} \| \sum_{i=1}^n a_i^T a_i \| & \|a_i\| - \|a_i\|^2 \in [0, \frac{1}{4}] \\
&= \frac{1}{4N} \|A^T A\| \\
&= \frac{1}{4N} \lambda_{\max}(A^T A) \\
&= \frac{1}{4N} \|Aw\|^2
\end{aligned}$$

□

Lemma 4.10. For all $N, t \geq 0$, $\Pr(\|Z\| \geq t/2) \leq \exp\left(\ln(2n) - \frac{Nt^2/8}{t/6 + \nu}\right)$, where $\nu = \|Aw\|^2/4$

Proof. The first line of the following proof holds by Matrix Bernstein's Inequality (5.4.1 in Vershyn's HDP), and Lemmas 4.1, 4.6, 4.7, and 4.9

$$\begin{aligned}
\Pr(|Z| \geq t/2) &\leq 2n \exp\left(-\frac{t^2/8}{\frac{t}{6N} + \frac{1}{N}\nu}\right) \\
&= 2n \exp\left(-\frac{Nt^2/8}{t/6 + \nu}\right) \\
&= \exp(\ln(2n) - \frac{Nt^2/8}{t/6 + \nu})
\end{aligned}$$

□

Lemma 4.11. For all $N, t \geq 0$, $\Pr\left(\left||A\tilde{w}|^2 - |Aw|^2\right| \geq t\right) \leq \exp\left(\ln(2n) - \frac{Nt^2/8}{t/6 + \nu}\right)$, where $\nu = |Aw|^2/4$

Proof.

$$\begin{aligned}
\Pr\left(\left||A\tilde{w}|^2 - |Aw|^2\right| \geq t\right) &\leq \Pr(2|Z| \geq t) && \textbf{Lemma 4.5} \\
&\leq \Pr(|Z| \geq t/2) \\
&\leq \exp(\ln(2n) - \frac{Nt^2/8}{t/6 + \nu}) && \textbf{Lemma 4.10}
\end{aligned}$$

□

Lemma 4.12. For all $N > 0$, $\epsilon \in [0, 1]$, with probability at least $1 - \exp\left(\ln(2n) - \frac{N||Aw||^2(2\epsilon - \epsilon^2)^2/4}{(2\epsilon - \epsilon^2)/3 + 1/2}\right)$, $||A\tilde{w}|| \in [(1 - \epsilon)||Aw||, ||Aw||]$

Notice that if $||Aw||$ and ϵ are constants, then the probability approaches 1 as n approaches infinity, so long as $N = \omega(\log n)$

Proof. By Lemma 4.11, with probability at least $1 - \exp\left(\ln(2n) - \frac{Nt^2/8}{t/6 + \nu}\right)$,

$$\left||A\tilde{w}|^2 - |Aw|^2\right| \leq t$$

Let $t = ||Aw||^2(2\epsilon - \epsilon^2)$, then, with probability at least $1 - \exp\left(\ln(2n) - \frac{N||Aw||^4(2\epsilon - \epsilon^2)^2/8}{||Aw||^2(2\epsilon - \epsilon^2)/6 + \nu}\right) \dots$

$$\left||A\tilde{w}|^2 - |Aw|^2\right| \leq ||Aw||^2(2\epsilon - \epsilon^2)$$

Notice first that the probability can be simplified since $\nu = ||Aw||^2/4$, thus the probability that $\left||A\tilde{w}|^2 - |Aw|^2\right| \leq ||Aw||^2(2\epsilon - \epsilon^2)$, holds is at least ...

$$1 - \exp\left(\ln(2n) - \frac{N||Aw||^2(2\epsilon - \epsilon^2)^2/4}{(2\epsilon - \epsilon^2)/3 + 1/2}\right)$$

Note that by operator norm definition, $||A\tilde{w}|^2 \leq ||Aw|^2$, so the lower bound is what is of significance, i.e.

$$\begin{aligned}
&-||Aw||^2(2\epsilon - \epsilon^2) \leq ||A\tilde{w}|^2 - |Aw|^2 \\
\iff &(1 - 2\epsilon - \epsilon^2)||Aw||^2 \leq ||A\tilde{w}|^2 \\
\iff &(1 - \epsilon^2)^2||Aw||^2 \leq ||A\tilde{w}|^2 \\
\iff &(1 - \epsilon^2)||Aw|| \leq ||A\tilde{w}|
\end{aligned}$$

$\therefore$ With probability at least $1 - \exp\left(\ln(2n) - \frac{N\|Aw\|^2(2\epsilon - \epsilon^2)^2/4}{(2\epsilon - \epsilon^2)/3 + 1/2}\right)$,

$$\|A\tilde{w}\| \in [(1 - \epsilon)\|Aw\|, \|Aw\|]$$

□

Lemma 4.13. For all $N > 0$, $\epsilon \in [0, 1]$, with probability at least $1 - \exp\left(\ln(2n) - \frac{1}{4}N\|Aw\|^2\epsilon^2\right)$, $\|A\tilde{w}\| \in [(1 - \epsilon)\|Aw\|, \|Aw\|]$

Proof. By Lemma 4.12 $\|A\tilde{w}\| \in [(1 - \epsilon)\|Aw\|, \|Aw\|]$ holds with probability at least ...

$$\begin{aligned} 1 - \exp\left(\ln(2n) - \frac{N\|Aw\|^2(2\epsilon - \epsilon^2)^2/4}{(2\epsilon - \epsilon^2)/3 + 1/2}\right) &\geq 1 - \exp\left(\ln(2n) - \frac{1}{4}N\|Aw\|^2\epsilon^2(2 - \epsilon)^2\right) \quad \forall x \in \mathbb{R}, (2x - x^2)/3 + 1/2 \leq 1 \\ &\geq 1 - \exp\left(\ln(2n) - \frac{1}{4}N\|Aw\|^2\epsilon^2\right) \quad \forall x \in [0, 1], x^2(2 - x)^2 \leq x^2 \end{aligned}$$

□

Lemma 4.14. For all $N > 0$, $\delta \in [0, 1]$, with probability at least δ , $\|A\tilde{w}\| \in [(1 - \epsilon)\|Aw\|, \|Aw\|]$, where $\epsilon = 1 - \sqrt{1 - \frac{2}{3} \cdot \frac{L + \sqrt{L(L + 9N\|Aw\|^2/2)}}{N\|Aw\|^2}}$, where $L = \ln(\frac{2n}{1 - \delta})$

Notice that if $\|Aw\|$ and δ are constants, then $\epsilon = O(\sqrt{\frac{\log n}{N}} + \sqrt{\frac{\sqrt{N \log n}}{N}})$, and if $N = \omega(\log(n))$, then as n approaches infinity, ϵ approaches zero.

Proof. Let $\eta = 2\epsilon - \epsilon^2$

$$\begin{aligned}
\delta &= 1 - \exp\left(\ln(2n) - \frac{N||Aw||^2\eta^2/4}{\eta/3 + 1/2}\right) \\
\iff 1 - \delta &= \exp\left(\ln(2n) - \frac{N||Aw||^2\eta^2/4}{\eta/3 + 1/2}\right) \\
\iff \ln(1 - \delta) &= \ln(2n) - \frac{N||Aw||^2\eta^2/4}{\eta/3 + 1/2} \\
\iff \ln(2n) - \ln(1 - \delta) &= \frac{N||Aw||^2\eta^2/4}{\eta/3 + 1/2} \\
\iff L &= \frac{N||Aw||^2\eta^2/4}{\eta/3 + 1/2} \\
\iff \frac{L}{3}\eta + L/2 &= N||Aw||^2\eta^2/4 \\
\iff 0 &= \frac{N||Aw||^2}{4}\eta^2 - \frac{L}{3}\eta - \frac{L}{2} \\
\iff \eta &= \frac{\frac{L}{3} \pm \sqrt{\frac{L^2}{9} + 4 \cdot \frac{N||Aw||^2}{4} \cdot \frac{L}{2}}}{2 \frac{N||Aw||^2}{4}} \\
\iff \eta &= \frac{2}{3} \frac{L \pm 3\sqrt{L(\frac{1}{9}L + \frac{1}{2}N||Aw||^2)}}{N||Aw||^2} \\
\iff 2\epsilon - \epsilon^2 &= \frac{2}{3} \frac{L + \sqrt{L(L + \frac{9}{2}N||Aw||^2)}}{N||Aw||^2} \\
\iff 0 &= \epsilon^2 - 2\epsilon + \frac{2}{3} \frac{L + \sqrt{L(L + \frac{9}{2}N||Aw||^2)}}{N||Aw||^2} \\
\iff \epsilon &= 1 - \sqrt{1 - \frac{2}{3} \frac{L + \sqrt{L(L + \frac{9}{2}N||Aw||^2)}}{N||Aw||^2}}
\end{aligned}$$

By Lemma 4.12, with probability at least, δ , $||A\tilde{w}|| \in [(1 - \epsilon)||Aw||, ||Aw||]$. □

4.1 Expected Sampling Amount

The goal of this section is to determine the expected amount of sampling that occurs.

Let Z be the random variable, that is the total number of nonzero rows of N iid copies of $\tilde{A}_i$, i.e. $Z = \sum_{i=1}^N \sum_{j=1}^n X_{ij}$.

Lemma 4.15. $\mathbb{E}[Z] = \sum_{i=1}^N \sum_{j=1}^n ||a_j||$.

Proof.

$$\begin{aligned}\mathbb{E}[Z] &= \sum_{i=1}^N \sum_{j=1}^n \mathbb{E}[X_{ij}] \\ &= \sum_{i=1}^N \sum_{j=1}^n ||a_j||\end{aligned}$$

LOE

□

Lemma 4.16. $\mathbb{E}[Z] \leq \begin{cases} cN(1 + \frac{n^{-k+1}-1}{-k+1}) & k \neq 1 \\ cN(1 + \ln(n)) & k = 1 \end{cases}, \text{ where } c > 0$

Proof.

$$\begin{aligned}\mathbb{E}[Z] &= \sum_{i=1}^N \sum_{j=1}^n ||a_j|| \\ &\leq c \sum_{i=1}^N \sum_{j=1}^n j^{-k} \\ &= cN \sum_{j=1}^n j^{-k} \\ &\leq cN(1 + \int_{x=1}^n x^{-k} dx) \\ &= \begin{cases} cN(1 + \frac{n^{-k+1}-1}{-k+1}) & k \neq 1 \\ cN(1 + \ln(n)) & k = 1 \end{cases}\end{aligned}$$

Lemma 4.15

WLOG $||a_j|| \leq cj^{-k}$

□

Lemma 4.17. $\mathbb{E}[Z] = \begin{cases} O(Nn^{1-k}) & \text{when } k \in (\frac{1}{3}, 1) \\ O(N \log n) & \text{when } k = 1 \\ O(N) & \text{when } k \in (1, \infty) \end{cases}$

Proof.

$$\begin{aligned}\mathbb{E}[Z] &\leq \begin{cases} cN(1 + \frac{n^{-k+1}-1}{-k+1}) & k \neq 1 \\ cN(1 + \ln(n)) & k = 1 \end{cases} \\ &= \begin{cases} O(Nn^{1-k}) & \text{when } k \in (\frac{1}{3}, 1) \\ O(N \log n) & \text{when } k = 1 \\ O(N - \frac{N}{n^{k-1}}) & \text{when } k \in (1, \infty) \end{cases} \\ &= \begin{cases} O(Nn^{1-k}) & \text{when } k \in (\frac{1}{3}, 1) \\ O(N \log n) & \text{when } k = 1 \\ O(N) & \text{when } k \in (1, \infty) \end{cases}\end{aligned}$$

Lemma 4.16

□